IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 18: EIA Case study-II

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering

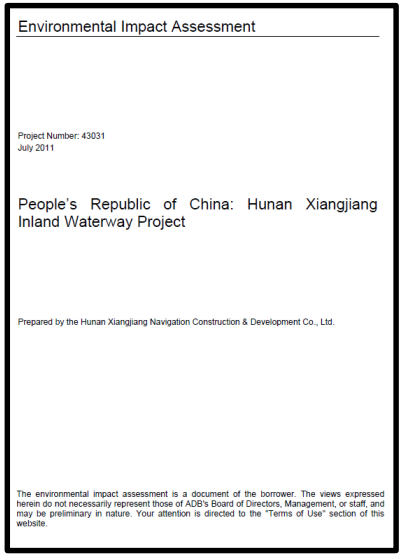


Contents

- EIA of an inland waterway project (Hunan Xiangjiang Inland Waterway)
  - Project details, location
  - Main component and benefits of project
  - EIA & environmental law regulations
  - Steps and process of EIA in China
  - Benefits over other modes
  - Location alternatives
  - Impact & mitigations
  - Public participation
  - Grievance redressal
  - Monitoring

Hunan Xiangjiang Inland Waterway Project

- Country: People's Republic of China
- Project type: construction of a multi-purpose river barrage, improve the navigational draft for shipping along a 50 km reach of the Xiangjiang River and generate hydropower
- Lender: Asian Development Bank (ADB)
- EIA report by: Hunan Xiangjiang Navigation Construction & Development Co., Ltd. Year 2011



Environmental Impact Assessment

Project Number: 43031
July 2011

People's Republic of China: Hunan Xiangjiang Inland Waterway Project

Prepared by the Hunan Xiangjiang Navigation Construction & Development Co., Ltd.

The environmental impact assessment is a document of the borrower. The views expressed herein do not necessarily represent those of ADB's Board of Directors, Management, or staff, and may be preliminary in nature. Your attention is directed to the "Terms of Use" section of this website.

Asian Development Bank (ADB)

- The Asian Development Bank (ADB) envisions a prosperous, inclusive, resilient, and sustainable Asia and the Pacific
- Assists its members and partners by providing loans, technical assistance, grants, and equity investments to promote social and economic development
- 68 member countries including India



Image: Logo of ADB

Project location

- Located in the middle to lower reaches of the Xiangjiang River, a tributary of the Yangtze River which flows from south to north, joining the Yangtze on its right-hand bank near the Chenglingji.

- Proposed Tugutang Complex is located approximately 38 km upstream from Hengyang municipality and about 50 km downstream from the Jinweizhou Complex.

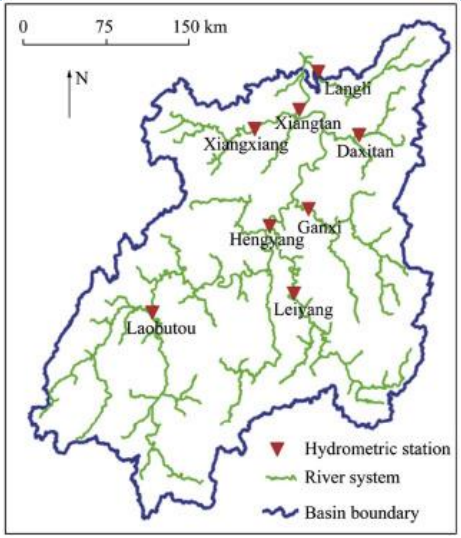


Image: Xiangjiang river basin

Image Source: Wang G. et al. 2017

Main component of project

- Construction of the Tugutang river barrage to raise water levels and flood dangerous shoals along a 50 km reach.
- Construct a hydropower station as part of the barrage with a total installed capacity of 90MW and normal annual energy output of 358.2 MkWh.
- Provide two new river wharves for handling goods for export downstream and import upstream.
- Undertake various works in the areas affected by inundation including resettlement of farmers and removal and replacement of structures and assets.

Need of the project 1/2

- In Hunan Province, the hilly area to the south and relatively narrow valley of the Xiangjiang River pose constraints to the development of both road and rail
- Navigation on the Xiangjiang River is poor
- Total industrial output along the river accounts for 60% of the whole province



Image: proposed site of project

Need of the project 2/2

- Volume of freight predicted to grow to 0.20 and 0.29 billion tons by the years 2020 and 2030 respectively from 20 million tons in 2010
- Lowering of transport costs and increasing the competitiveness of inland waterways against other forms of long-distance travel
- Energy consumption of unit transport is about 65% of railways and 15% of highway transport
- CO2 emissions from water transport are less than half those for rail and 22% of those for highway transport



Image: proposed site of project

Arrangement For The Tugutang river barrage



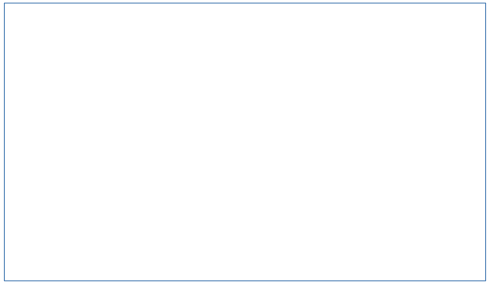
Image: Proposed barrage



Image: Barrage after construction

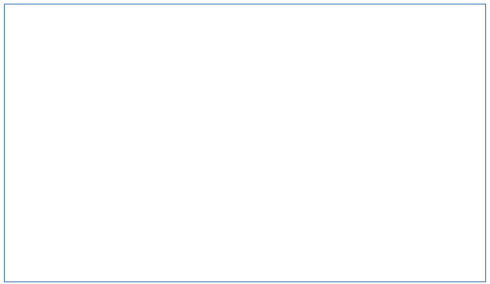
EIA requirement of Asian Development Project (ADB)

- The ADB EIA requirements are set out in Safeguard Policy Statement (2009). Like World bank EIA requirement & process.
- This Project has been screened for expected environmental impacts and categorized as Category A
- Projects with potential for significant adverse environmental impacts
- An environmental impact assessment is required to address significant impacts



Major environmental laws & regulations applicable

Year	Law
1989	Environment protection law
1997	Prevention & control of environmental noise pollution
1998	Protection of wildlife
1998	Flood control law
2001	Evaluation of environmental effects
2002	Water law
2004	Fishery law
2005	Prevention & control of pollution by solid waste
2008	Prevention & control of water pollution
2009	Prevention & control of atmospheric pollution



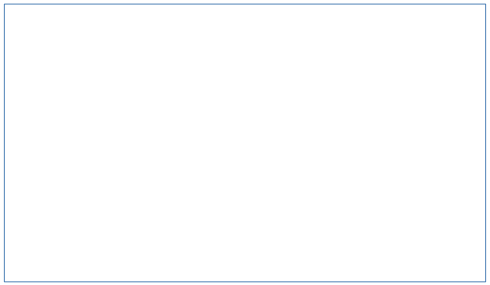
Steps in EIA process of China

- Environmental baseline data
- Project alternatives
- Recommendations for minimizing, mitigation and elimination of impacts
- Analysis of the extent and distribution of adverse impacts and their consequences
- Opinions of affected persons (public participation)
- Environmental monitoring program

Comparison with ‘No-Project’ scenario

In case of ‘No-Project’:

- River would not be navigable by ships during the dry season between the lower and upper sections.
- Cargo would have to be loaded off ships and onto road or rail freight and transported by land between Jinweizhou and Hengyang City (for 88 Km.).



Advantage of Inland Water Transport (IWT) over other modes

- Geography of the Xiangjiang River Corridor is well suited to IWT
- Lower operating costs per ton-km than compared to both road and rail
- IWT does not experience road congestion or suffer from the capacity issues of the rail freight system in the Province
- Relatively safe and secure for the shipment of goods

Location alternatives for barrage: Technical aspect

Items	Upstream	Middle	Downstream	Evaluation
Effectiveness of the Project				
Engineering feasibility	Suitable geological conditions exist. Topography more suitable than other sites.	Suitable geological conditions exist	Suitable geological conditions exist	Upstream preferred
Navigation conditions	River section is straight – better for navigation entry and exit to Locks. River depth 1.0~1.8m during lower water period.	Slight meandering river; upstream of dam site is 90° turning section. River level affected by downstream Dayuandu reservoir, 2~3.5m in depth.	Slightly meandering river 1km downstream dam site is 90° turning section. Dayuandu dead water depth is 2~3.5m.	Upstream preferred
Investment cost	Comparable to Middle site.	Comparable to Upstream site	Higher than Upstream and Middle sites due to higher height of barrage structure.	Upstream or Middle preferred
Hydropower generation	Lower power generation than Middle and Downstream sites	Comparable to Downstream site	Comparable to Middle site.	Middle or Downstream preferred
Compatibility with Masterplan Strategy	Compatible.	Compatible.	Conflicts with already built Dyuandu complex, and would reduce power generation there	Downstream not viable

- Green shade: Positive
- Red shade: Negative

Location alternatives for barrage: Environmental aspect 1/2

Attributes	Upstream	Middle	Downstream
• Excavation (million Meter cube)	• 1.8	• 2.9	• 2.1
• Water quality	Upstream of major wastewater outfalls and waterworks inlets	Urban and industrial wastewater outfalls within the reservoir area	• Urban and industrial wastewater outfalls within the reservoir area • Water intake of Hengnan County Waterworks is within the reservoir area

- Green shade: Positive

Location alternatives for barrage: Environmental aspect 2/2

Attributes	Upstream	Middle	Downstream
• Aquatic eco system	• Barrage coincides with fish spawning areas	• Same as upstream	• Same as upstream
• Dredging work	• Major dredging required downstream to connect with dead level of Dayuandu reservoir	• Minor dredging required to connect with dead level of Dayuandu reservoir	• No dredging required to connect with dead level of Dayuandu reservoir.

- Green shade: Positive

Location alternatives for barrage: Social aspect 1/2

Attribute	Upstream	Middle	Downstream
• Land	• Inundates 4 thousand mu (unit of area) of cultivated land	• Inundates 4½ thousand mu of cultivated land	• Inundates 5 thousand mu of cultivated land
• Resettlement	• Same people as other two options	• -	• -

- Green shed: Positive
- 1 mu= 6.07 acre

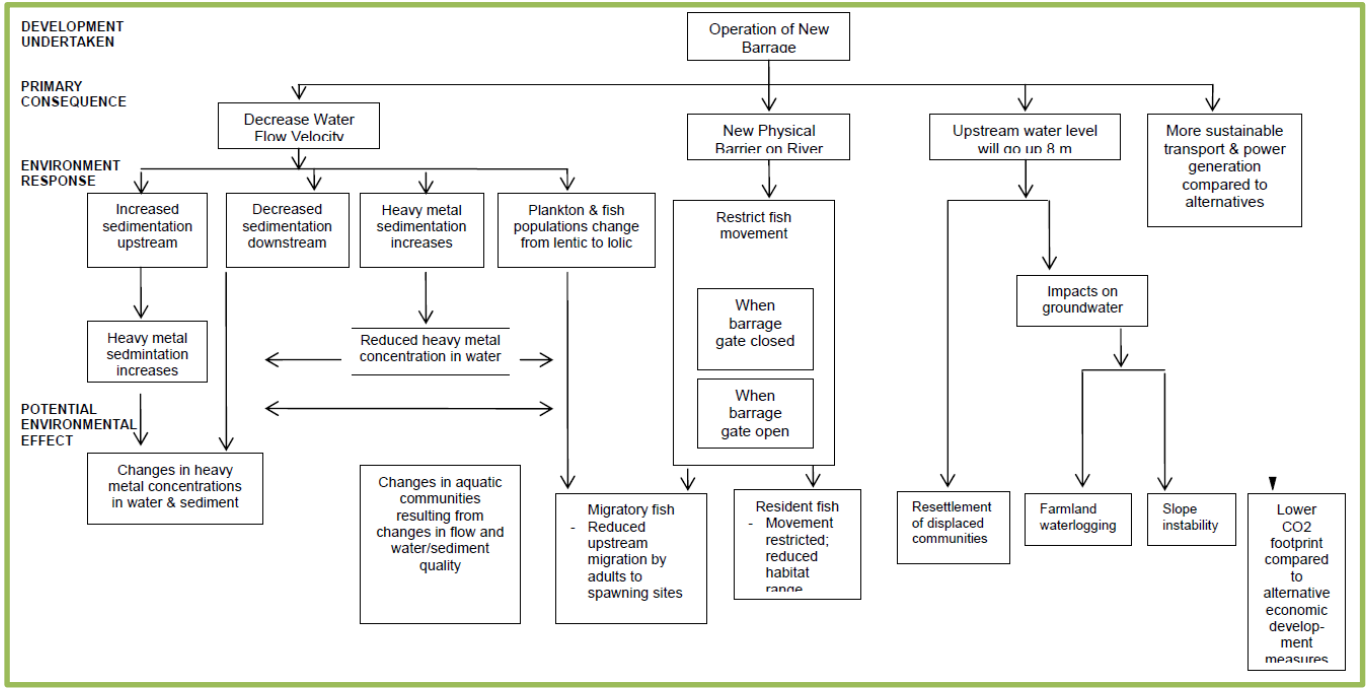
Location alternatives for barrage: Social aspect 2/2

Attribute	Upstream	Middle	Downstream
• Flood & drainage	• No impact on Yunji town	• Adjacent to Yunji town.	• Adjacent to development downstream of Yunji town.
• Resettle-ment	• Relatively few adjacent residential areas	• Site potentially in conflict with Hengan County industrial park and proposed Hengyang airport	• Left bank is adjacent to Chejiang town, with densely populated residential area

- Green shed: Positive

Upstream location was selected.

Overview of scoping of project's impacts



Impacts of project

Environment resource		Project action: Construction period: Land occupation	Vegetation removal	Stockpiles	Dredging	Machinery operation	Materials transport and placement	Concrete batching plant	Stone quarries	Initial filling of the reservoir	Operation period: Reservoir impoundment	Transportation	Energy generation	Vegetation recovery
Social Development	Employment		■	Δ	Δ						Δ		□	
	Navigation										□			
	Economic growth										□	□	□	
	Flood control				▲						Δ			
	Travel										□			□
	Land use	■	■	▲					■	▲	■			□
Material Resources	Hydrology				▲					▲	■			
	Water quality			▲	▲		▲	▲		▲	■			
	Air quality			▲	▲	▲	▲	▲				■		□
	Acoustic environment			▲	▲	▲	▲	▲				■		□
	Soil erosion	▲		▲	▲				▲	▲				□
Ecological Resources	Aquatic									▲	■			
	Vegetation	■		■					▲					□
	Terrestrial organisms	▲		▲										□
Life Quality	Landscape			▲	▲			▲			□			□
	Residence		□											□
	Traffic				▲		▲				□	□		
Remarks		□ Long term beneficial impact Δ Short term beneficial impact ■ Long term negative impact ▲ Short term negative impact												

Source: HydroChina Zhongnan Engineering Corporation EIA.

Impact on hydrology & mitigation measures

Impact	Mitigation
• Flow change due to the installation of coffer dams during construction • Decreased flows downstream of the barrage during initial impoundment. • Change in flow characteristics from river to a reservoir • Changes in flow downstream of the barrage • Decreases in velocity around the new river wharfs	• Coffer dams will be temporary • filling of the reservoir will ideally be programmed for the wet season (minimum disturbance to flow) • Programmed to modify the hydrology of the river for specific purposes like, to minimize the risk of flooding during extreme storm or runoff events

Impact on water quality & mitigation measures

Impact	Mitigation
• Discharge of sewage and wastewaters from construction sites and camps to surface waters • Risk of accidental spillages of oils, fuels, and other materials • Lower diffusion and transportation of pollutants	• Collection of construction water, groundwater pumped from excavations, and stormwater runoff in sedimentation ponds prior to discharge to the river • Two bio-filter sewage treatment plants, one on each side of the river, to treat domestic wastewater • Treatment of oily water from the workshops • Sedimentation ponds for water used in aggregate washing

Impact on ground water level & mitigation measures

Impact	Mitigation
• Groundwater levels in the surrounding area are expected to rise by 0.5 to 2.1m depending on location and soil profile	• Flooding will be mitigated by removing soils from nearby hilltops and using it to raise land in affected areas

Impact on fisheries & mitigation measures

Impact	Mitigation
• Changes in the eco-environment and their effects on fish • Impact on rare and specific fish resources • Impact on fish in Dongting Lake • Impacts on important eco-environments of fish	• Fishing prohibited during the spawning season • Allow movement of migratory fish by coordinating the opening of sluice gates on the barrages during the floods • Construction of a fish pass alongside the barrage • Construction of a fish hatchery and implementation of a long-term programme for artificial releases of fish spawn into the river.

Impact on air quality & mitigation measures

Impact	Mitigation
• Dust & emission during construction • Air pollution during operation would be the emissions (mostly SO2 and NO2)	• implement Dust and Emissions control programme to control the generation and dispersion of dust and vehicle emissions • Air quality monitoring would be conducted during construction to monitor compliance against the environmental quality standards.

Mitigation measures for displaced population

- Compensation for permanent land loss comprising a compensation fee for loss of land, a resettlement subsidy and compensation for the loss of standing crop.
- Compensation for temporary land loss, including loss of standing crop and restoration fee.
- Compensation for loss / relocation of houses and structures, and relocation allowance, crop losses.
- Compensation for loss of business / employment.
- Compensation for public facilities.

Public participation: EIA hearing forum

- A hearing forum on the EIA of the project was held on August 4, 2010
- Project details were shared
- Most of the hearing representatives supported the project
- Queries from public were answered





Public participation: Group investigation





- The survey of groups using the questionnaire was completed
- 300 questionnaires were handed out and all were completed, a response rate of 100%.
- 95% of the residents agree that the project is conducive to regional economic development

Public participation: Web publicity

- Published information about the project EIA on the government website of Hengnan County also on website of the Immigration Bureau of Changning City
- Internet publicity was posted for at least 10 days
- Views were received from the People's Congress of Hengnan County.
- Main comment was that scientific comparison and verification should be made for choosing the best programme

Public participation: Field publicity-first round

- Posted public notices in the region affected by the project
- Notices provided basic information about the project and preliminary environmental impact analysis
- Informed people on how to provide their views on environmental impact



Public participation: Field publicity- second round

- Second round of public notices was posted at the same locations, updating the information previously posted
- The notices included information about the project, potential environmental impacts, mitigation measures

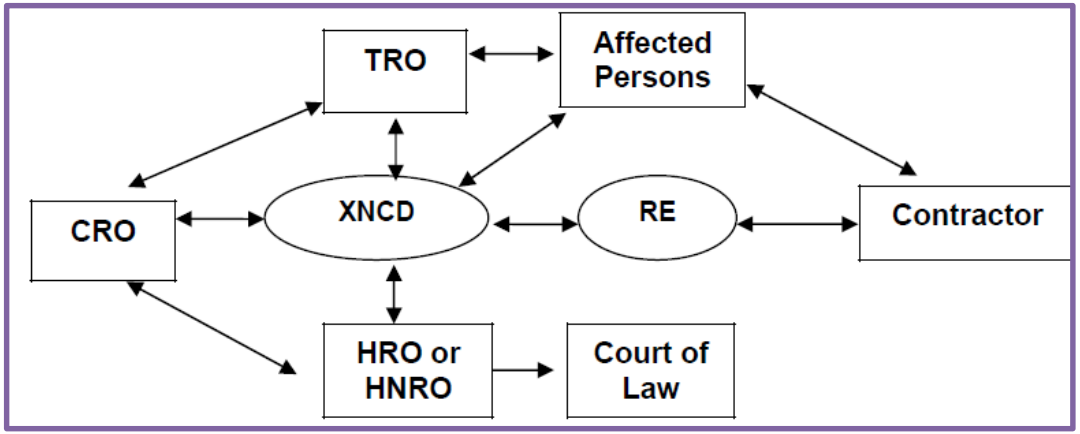




Public concerns & redressal

- Concerns concentrated mainly on land acquisition and resettlement and environmental impact.
- Local people who disapproved of the project were mainly afraid that the land requisition and demolition would not be in place and that compensation would be less, which would be detrimental to the interests of the individual.
- In response, the construction organizations promised that the compensation would be paid strictly according to the relevant national provisions on resettlement and the interests of residents would not be harmed.

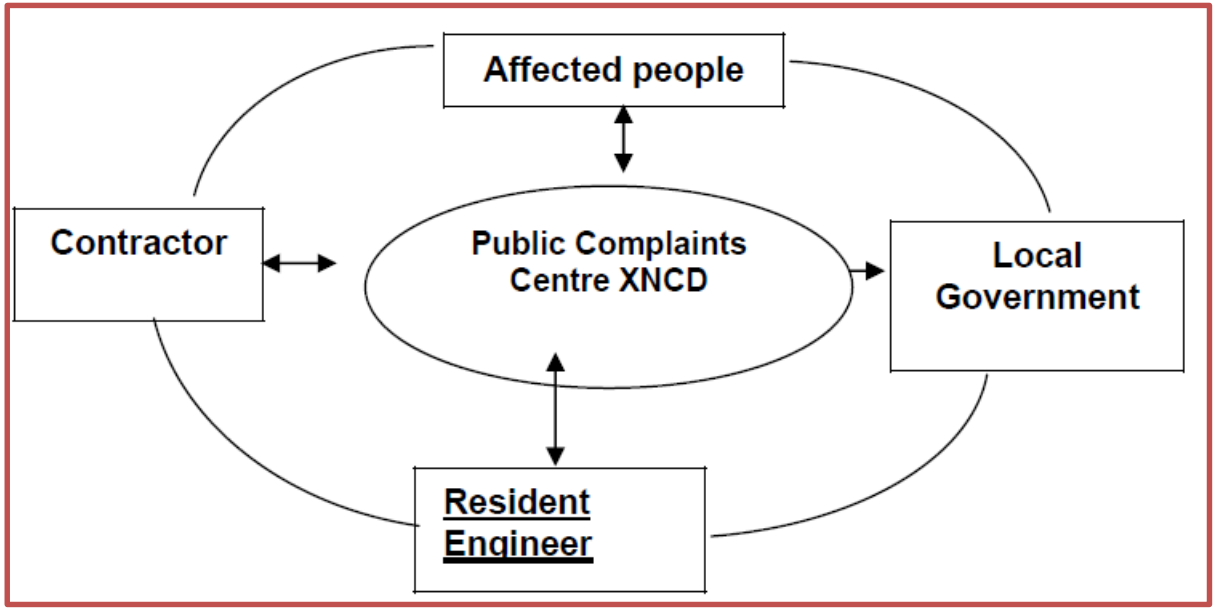
Grievance redressal mechanism: Old mechanism



- TRO: Township Resettlement Officer
- CRO: City Resettlement Officer
- HRO: Hunan (Province) Resettlement Officer
- XNCD: State council for development
- RE: Resident Engineer

- Old mechanism in practice was complicated for general public and it had lack of clarity

Grievance redressal Mechanism: New simplified mechanism



- New mechanism was adopted for this project
- Simple & clear in understanding

Monitoring during construction phase 1/2

- Wastewater quality from the excavations for the foundations; sand washing, concrete mixing and domestic sewage treatment plant.
- River water quality: upstream and downstream of the dam site.
- Air quality: quarterly monitoring at three sites near villages.
- Noise: monitoring at sensitive locations around the Tugutang Dam site on one day a month over 12 occasions.

Monitoring during construction phase 2/2

- Terrestrial Ecosystems: Vegetation cover and crops at and around construction sites.
- Population health: Health checks on 10% of the construction workforce followed by quarterly reporting on infectious diseases.
- Soil and water conservation: Monitoring in the wet season.

Monitoring during operation 1/2

- Wastewater arising from domestic sewage at the offices of the Tugutang Complex and ports.
- River water quality: At four locations in the Tugutang reach.
- Sediments: Heavy metals in the riverbed sediments Tugutang reach.
- Aquatic ecology: Monitoring hydrologic and hydrodynamic characteristics of the flow, plankton, benthos and fisheries.

Monitoring during operation 2/2

- Terrestrial ecosystems: Three monitoring campaigns pre-construction, and two after construction, in the spring and the autumn at selected sites around the reservoir.
- Population Health: General survey of the health of the population in the reservoir and resettlement areas.
- Soil and Water Conservation: Monitoring on success of vegetation recovery and replacement of failed planting areas.

Conclusion

- Project was important for interconnectivity through inland water transport.
- Beneficial over other modes of transport.
- Impact were assessed and mitigation measures were adopted for these adverse impact.
- A holistic public participation was done.
- Monitoring of vital environmental elements is in place.

References:

- Asian Development Bank, “ Environment impact assessment: People’s Republic of China: Hunan Xiangjiang Inland Waterway Project”, project no. 43031; 2011

- Guo-qing Wang, Jian-yun Zhang, Yue-ping Xu, Zhen-xin Bao, Xin-yue Yang, Estimation of future water resources of Xiangjiang River Basin with VIC model under multiple climate scenarios, Water Science and Engineering, Volume 10, Issue 2 ;2017

Thank You